Use the following values for resistivities

Table 28.1 RESISTIVITIES OF METALS[a]

Material	ρ	*Increase in ρ per °C*
Silver	$1.6 \times 10^{-8}\ \Omega \cdot m$	0.38%
Copper	1.7×10^{-8}	0.39%
Aluminum	2.8×10^{-8}	0.39%
Brass	$\sim 7 \times 10^{-8}$	0.2%
Nickel	7.8×10^{-8}	0.6%
Iron	10×10^{-8}	0.5%
Steel	$\sim 11 \times 10^{-8}$	0.4%
Constantan	49×10^{-8}	0.001%
Nichrome	100×10^{-8}	0.04%

[a] At a temperature of 20°C.

*16. A lightning rod of iron has a diameter of 0.80 cm and a length of 0.50 m. During a lightning stroke, it carries a current of 1.0×10^4 A. What is the potential drop along the rod?

10. The collision time for electrons in the metal sodium (atomic symbol Na) is 8.8×10^{-15} s at room temperature. Assuming one free electron per atom, calculate the electrical resistivity of sodium.

12. A high-voltage transmission line has an aluminum cable of diameter 3.0 cm, 200 km long. What is the resistance of this cable?

*13. In silver, the number of free electrons is $n = 5.8 \times 10^{28}$ per cubic meter.
(a) Using the value of the resistivity given in Table 28.1, calculate the average time interval between collisions of one of these electrons.
(b) Assuming that the electric field in a current-carrying silver wire is 8.0 V/m, calculate the average drift velocity of an electron.

*33. A water pipe is made of iron with an outside diameter of 2.5 cm and an inside diameter of 2.0 cm. The pipe is used to ground an electric appliance. If a current of 20 A flows from the appliance into the water pipe, what fraction of this current will flow in the iron? What fraction in the water? Assume that water has a resistivity of 0.01 $\Omega \cdot$ m.

**29. A solid truncated cone is made of a material of resistivity ρ (Figure 28.15). The cone has a height h, a radius a at one end, and a radius b at the other end. Derive a formula for the resistance of this cone.

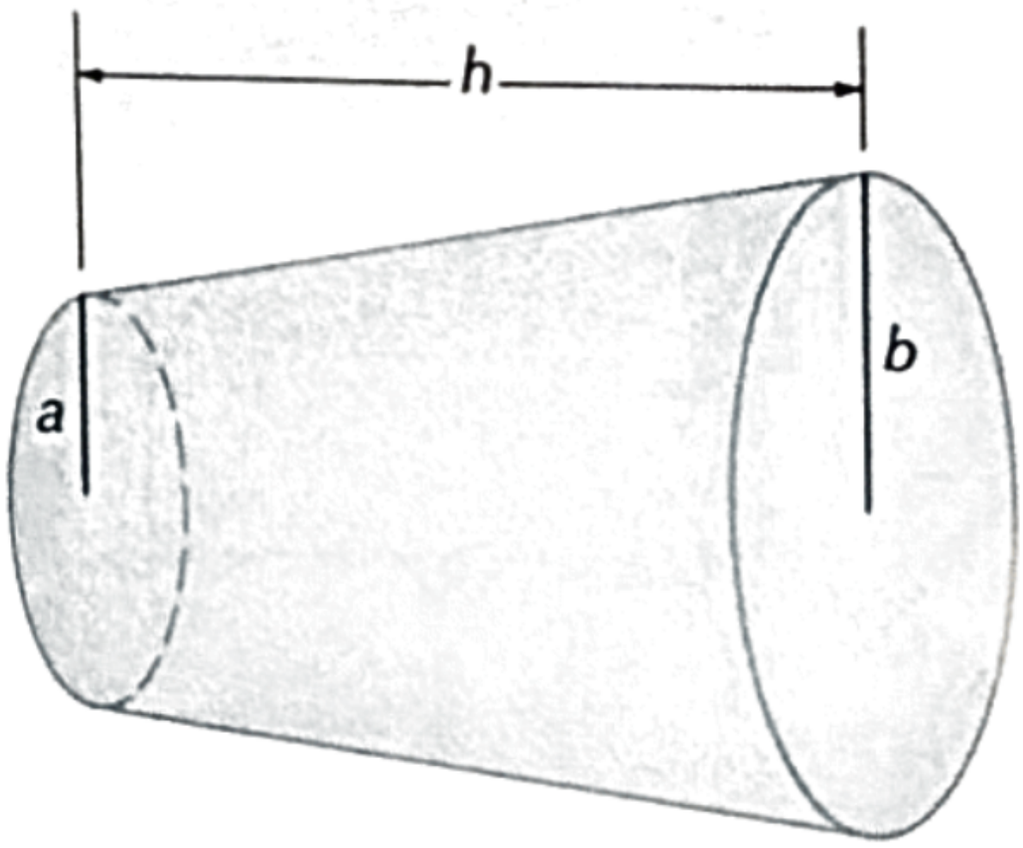


Fig. 28.15

AMT M **1.** A 200-km-long high-voltage transmission line 2.00 cm in diameter carries a steady current of 1 000 A. If the conductor is copper with a free charge density of 8.50×10^{28} electrons per cubic meter, how many years does it take one electron to travel the full length of the cable?

43. A 100-W lightbulb connected to a 120-V source experiences a voltage surge that produces 140 V for a moment. By what percentage does its power output increase? Assume its resistance does not change.